Crosspoints:

In the implementation, I defined two types of **CrossPoints**:

- **Positive CrossPoint:** A Magnetoresistive CrossPoint
- **Negative CrossPoint:** A Non-Magnetoresistive CrossPoint.

Magnetoresistive Crosspoint
state: index of x
- $G_p = a * log_{10}(x) + b$
- $G_{ap} = G_p(1 + c(G_p - G_{threshold})^{3/4})$
[based on previous version which can be modified]

Non Magnetoresistive Crosspoint
state: index of x
- $G_{bias} = a * log_{10}(x) + b$

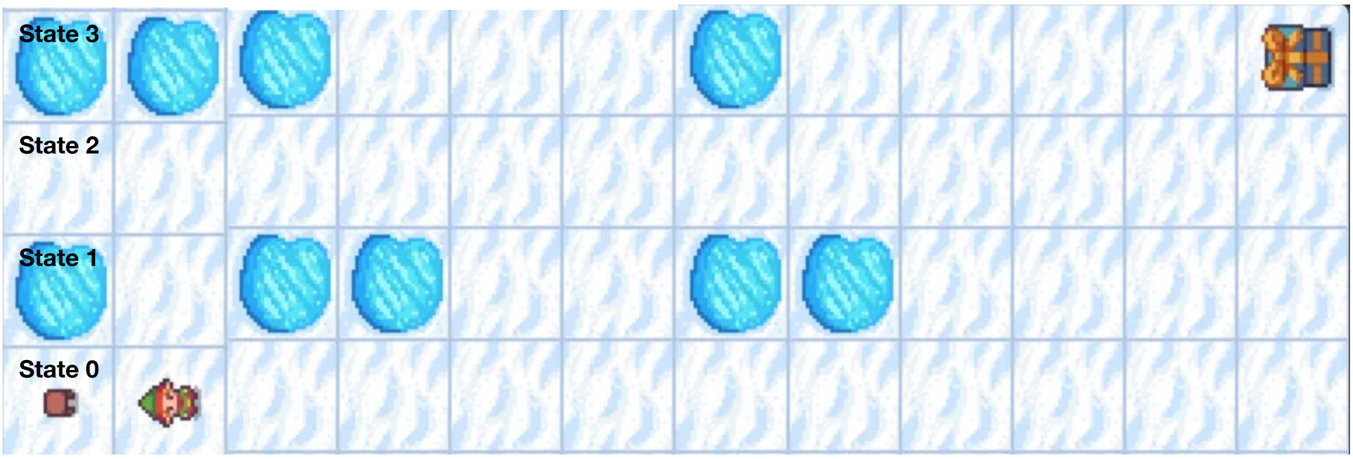
Synapse:

I then defined a **MultiWeightSynapse**, which consists of multiple Magnetoresistive CrossPoints and a single Non-Magnetoresistive CrossPoint. The weight calculation is implemented in the this class. It gets ap_index to specify which magnetoresistive crosspoint should be ap and others will be p, then weight is calculated for each synapse.

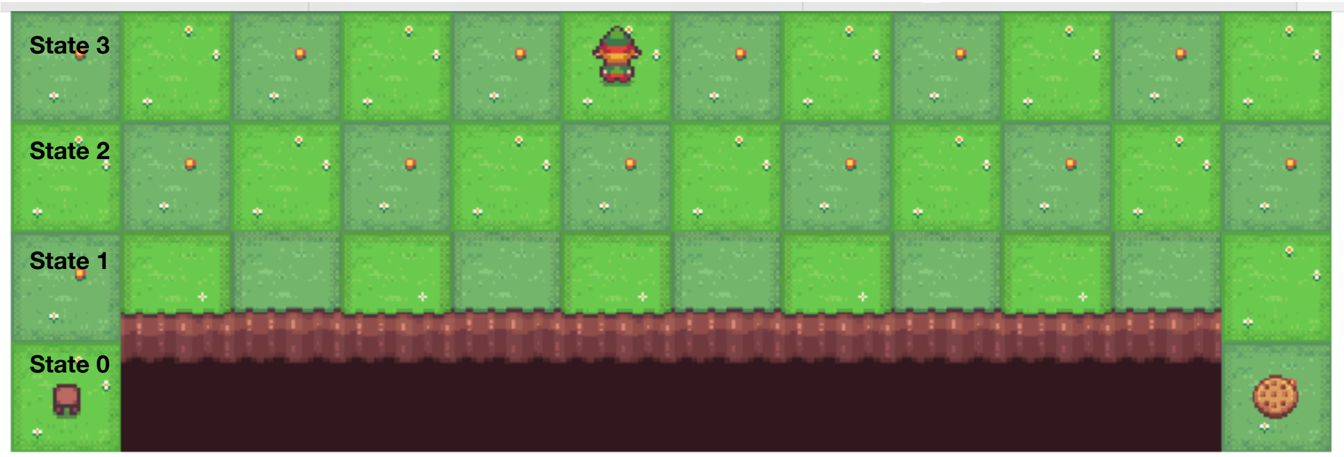
Multi Weight Synapse
- Two Magnetoresistive Crosspoint (ap=0, ap=1)
- One Non Magnetoresistive Crosspoint (bias)
- **weight(ap_index)**: according to ap returns: e.g. $ScalingFactor * (G_{0,P} + G_{1,AP} - G_{bias})$

Environments:

- Frozen Lake: contains an 4x12 states environment (48 grids) and 4 actions (up, down, left, right)



- Cliff Walking: contains an 4x12 states environment (48 grids) and 4 actions (up, down, left, right)



MemristiveQFunction:

The MemristiveQFunction is used to store action_values for the agent.(means how much is the value of action a in state s), each state–action pair is mapped to its own multi-weight synapse, resulting in a total of **48 × 4 synapses**.

Consider the agent is in state s = 10. For this state, the MemristiveQFunction contains four multi-weight synapses, one for each action (up, down, left, right).

If agent selects action “right” in state “10”, Q(state=10, action=right) is gained by reading the conductance of the corresponding synapse.

During training, if the learning rule shows value of taking action right in state 10 should be:

- increased, only the multi-weight synapse associated with the pair (state=10, action=right) is updated. The update is applied to a specific positive crosspoint selected by the `ap_index`, which indicates the active problem.

- decreased, the single bias’s index increased which affects both problems.